

UNIT-REGION FACTORIZATION FOR EMPTY-REGION PROXIMITY GRAPHS

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ABSTRACT. Many proximity graphs join two sites when a region determined by the pair contains no other site. We record the elementary volume factor that appears in the homogeneous Poisson calculations for such graphs. If the pairwise region is a similarity copy of one normalized Borel set K , that is,

$$\ell_{p,q} := \|p - q\|, \quad R_{p,q}e_1 = (q - p)/\ell_{p,q}, \quad S(p, q) = p + \ell_{p,q}R_{p,q}K,$$

then

$$\lambda_d(S(p, q)) = \ell_{p,q}^d a_K, \quad a_K := \lambda_d(K).$$

The fixed-pair Poisson void probability is then determined by a_K . For symmetric undirected rules, the same constant also gives the Palm incident-edge radial intensity, the normalized Palm edge-length law, the mean Palm degree, and the order- k empty-region analogue. Expected-size limits, stabilization results, normal approximations, and long-edge scales use this local scalar under the additional assumptions of the corresponding limit theorems. The examples treated below include Gabriel regions, relative-neighborhood lunes, fixed-parameter β -skeleton regions, finite empty-region templates of Cardinal-Collette-Langerman, finite Veltkamp-type ball regions, and the stepping-stone diversion region.

1. INTRODUCTION

Let $V \subset \mathbb{R}^d$ be finite or locally finite. An empty-region graph assigns a Borel set $S(p, q) \subset \mathbb{R}^d$ to each candidate pair $p \neq q$, and declares $\{p, q\}$ to be an edge when

$$S(p, q) \cap (V \setminus \{p, q\}) = \emptyset.$$

This pairwise-set formalism appears in Devroye's expected-size analysis of proximity graphs [5], in Cardinal-Collette-Langerman's empty-region graphs [4], and in Goldstein-Johnson-Lachière-Rey's proximity-region graphs [8]. Classical examples include Gabriel graphs, relative-neighborhood graphs, and β -skeletons [7, 14, 16, 9, 11].

We work with the template construction of Cardinal-Collette-Langerman: the region determined by a pair is obtained from a fixed normalized region by translation, rotation, and uniform scaling. If

$$\ell_{p,q} := \|p - q\|, \quad R_{p,q}e_1 = \frac{q - p}{\ell_{p,q}}, \quad S(p, q) = p + \ell_{p,q}R_{p,q}K,$$

then the volume calculation depends on the template only through

$$a_K := \lambda_d(K).$$

Indeed,

$$\lambda_d(S(p, q)) = a_K \ell_{p,q}^d.$$

Thus every homogeneous Poisson void calculation for such a rule contains the same scale

$$\rho a_K \ell_{p,q}^d.$$

The exact consequences are the fixed-pair Poisson void law and the order- k empty-region analogue. For symmetric undirected rules, the same scalar also gives the Palm incident-edge radial law, the normalized Palm edge-length distribution, and the mean Palm degree. Expected-size limits, stabilization results, normal approximations, and long-edge asymptotics are cited later as settings where this local constant appears together with the additional hypotheses of the corresponding limit theorems.

2. NOTATION AND STANDING CONVENTIONS

Throughout the paper, $d \in \mathbb{N}$ is the ambient dimension, $\mathbb{R}^d$ is equipped with the Euclidean norm $\|\cdot\|$, and λ_d denotes d -dimensional Lebesgue measure. The first coordinate vector is

$$e_1 = (1, 0, \dots, 0) \in \mathbb{R}^d,$$

and $O(d)$ denotes the orthogonal group. For a set $A \subset \mathbb{R}^d$, ∂A denotes its boundary.

For $x \in \mathbb{R}^d$ and $r \geq 0$, let

$$B(x, r) := \{y \in \mathbb{R}^d : \|y - x\| \leq r\}$$

be the closed Euclidean ball. Its volume is

$$(2.1) \quad \lambda_d(B(x, r)) = \kappa_d r^d, \quad \kappa_d := \lambda_d(B(0, 1)) = \frac{\pi^{d/2}}{\Gamma(1 + d/2)},$$

where Γ is Euler's gamma function. We also use the convention $\kappa_0 = 1$, which is needed when a $(d - 1)$ -dimensional slice appears and $d = 1$.

For every candidate pair $p \neq q$, define

$$(2.2) \quad \ell_{p,q} := \|p - q\|, \quad u_{p,q} := \frac{q - p}{\ell_{p,q}}.$$

Thus $\ell_{p,q}$ is the pair length and $u_{p,q}$ is the unit vector pointing from p to q . When a rotation is needed, $R_{p,q} \in O(d)$ denotes an orthogonal map satisfying

$$R_{p,q}e_1 = u_{p,q}.$$

The affine map

$$T_{p,q}(z) := p + \ell_{p,q} R_{p,q} z$$

sends the normalized segment $[0, e_1]$ to $[p, q]$. Its inverse is

$$T_{p,q}^{-1}(z) = R_{p,q}^{-1} \frac{z - p}{\ell_{p,q}}.$$

The symbol $K \subset \mathbb{R}^d$ denotes a normalized unit region, and

$$a_K := \lambda_d(K)$$

denotes its unit volume. The standing finite-volume assumption for unit-region rules is

$$0 < a_K < \infty.$$

In the probabilistic sections, Φ denotes a homogeneous Poisson point process on $\mathbb{R}^d$ with intensity measure $\rho \lambda_d$, where $\rho > 0$. The symbols $\mathbb{P}$ and $\mathbb{E}$ denote probability and expectation. Palm probability and expectation at the origin are denoted by $\mathbb{P}^0$ and $\mathbb{E}^0$. Under the reduced Palm convention, sums over candidate neighbors of the typical point are written as sums over $\Phi \setminus \{0\}$. For the homogeneous Poisson process, the Mecke–Slivnyak theorem identifies the reduced Palm process with a homogeneous Poisson process of intensity $\rho \lambda_d$.

The set

$$\mathbb{N}_0 := \{0, 1, 2, \dots\}$$

is used for order- k empty-region rules. Boundary conventions are fixed in the deterministic graph definition: the empty region may be open, closed, or given by an interior. For Poisson or binomial input with absolutely continuous intensity, these conventions give the same void probabilities whenever

$$\lambda_d(\partial S(p, q)) = 0.$$

For deterministic point sets, the chosen boundary convention can change the edge set when data points lie on $\partial S(p, q)$.

3. UNIT REGIONS

Definition 3.1 (Unit region). *A pairwise empty-region rule S has a unit region if there is a Borel set $K \subset \mathbb{R}^d$ with $0 < \lambda_d(K) < \infty$ such that, for every $p \neq q$, there exists $R_{p,q} \in O(d)$ satisfying*

$$R_{p,q}e_1 = u_{p,q}$$

and

$$(3.1) \quad S(p, q) = p + \ell_{p,q} R_{p,q} K.$$

We write $a_K := \lambda_d(K)$.

The same condition can be expressed as a template-region construction with a fixed normalized region that is translated, rotated, and uniformly scaled according to the candidate pair [4]. It is also implied by set-level similarity equivariance.

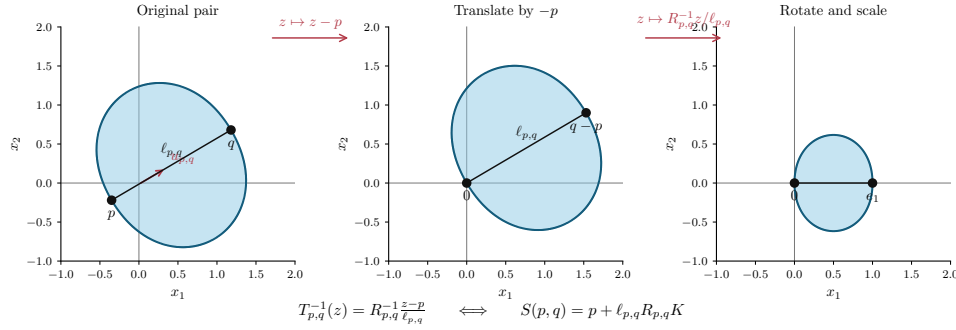


FIGURE 1. Normalization of a pairwise empty region. Starting from a candidate pair p, q , translate by $-p$, rotate the segment $q - p$ onto the e_1 -axis, and scale by $\ell_{p,q}^{-1}$. The normalized region is $K = S(0, e_1)$. The inverse map gives $S(p, q) = p + \ell_{p,q} R_{p,q} K$, and therefore $\lambda_d(S(p, q)) = \ell_{p,q}^d \lambda_d(K)$.

Proposition 3.2 (Similarity equivariance gives a unit region). *Suppose that for every Euclidean similarity $T(x) = \tau R x + a$, with $\tau > 0$, $R \in O(d)$, and $a \in \mathbb{R}^d$, one has*

$$S(Tp, Tq) = T(S(p, q)).$$

Then S has a unit region, namely $K = S(0, e_1)$.

Proof. For $p \neq q$, choose $R_{p,q} \in O(d)$ with $R_{p,q}e_1 = u_{p,q}$. The similarity $T(x) = p + \ell_{p,q} R_{p,q} x$ maps $(0, e_1)$ to (p, q) . Hence

$$S(p, q) = S(T0, Te_1) = T(S(0, e_1)) = p + \ell_{p,q} R_{p,q} K.$$

□

For $d \geq 3$, many orthogonal maps send e_1 to $u_{p,q}$. The volume identity is independent of this choice, since every orthogonal map preserves λ_d . The graph-level definition also needs a definite orientation convention when K lacks axial symmetry around the e_1 -axis. This can be supplied by a measurable choice $(p, q) \mapsto R_{p,q}$, or by imposing invariance of K under all rotations fixing e_1 . Symmetric Gabriel, lune, and stepping-stone regions satisfy this axial-symmetry condition.

Theorem 3.3 (Unit-region factorization). *If S has a unit region K , then*

$$(3.2) \quad \lambda_d(S(p, q)) = a_K \ell_{p,q}^d, \quad p \neq q.$$

Proof. Translation and orthogonal maps preserve Lebesgue measure. Scaling by $\ell_{p,q}$ multiplies d -dimensional volume by $\ell_{p,q}^d$. Applying this to (3.1) gives

$$\lambda_d(S(p, q)) = \lambda_d(p + \ell_{p,q} R_{p,q} K) = \ell_{p,q}^d \lambda_d(K).$$

□

4. POISSON VOID PROBABILITIES AND PALM CONSEQUENCES

Let Φ be a homogeneous Poisson point process on $\mathbb{R}^d$ with intensity measure $\rho\lambda_d$, where $\rho > 0$. Thus, for every finite-measure Borel set $A \subset \mathbb{R}^d$,

$$\Phi(A) \sim \text{Poisson}(\rho\lambda_d(A)),$$

and the counts on disjoint Borel sets are independent. In particular,

$$(4.1) \quad \mathbb{P}\{\Phi(A) = 0\} = \exp\{-\rho\lambda_d(A)\}.$$

This is the Poisson void probability [12]. Applying Theorem 3.3 to $A = S(p, q)$ gives the fixed-pair formula.

Corollary 4.1 (Fixed-pair Poisson void law). *If S has unit region K , then for every deterministic pair $p \neq q$,*

$$(4.2) \quad \mathbb{P}\{\Phi(S(p, q)) = 0\} = \exp\{-\rho a_K \ell_{p,q}^d\}.$$

We next pass from fixed deterministic pairs to the typical point of the Poisson process. The Palm statements below are for symmetric undirected empty-region rules. Under the Palm distribution at the origin, candidate neighbors are the points of the reduced process $\Phi \setminus \{0\}$. For a candidate point $x \neq 0$, the empty-region test is applied to the other Poisson points. By the Mecke–Slivnyak theorem, the reduced Palm process is again a homogeneous Poisson process with intensity $\rho\lambda_d$. Hence the Palm acceptance probability of a candidate at x is

$$\exp\{-\rho a_K \|x\|^d\}.$$

Define the Palm incident-edge length measure ν_K on $[0, \infty)$ by

$$\nu_K(C) := \mathbb{E}^0 \sum_{x \in \Phi \setminus \{0\}} \mathbf{1}\{0 \leftrightarrow x\} \mathbf{1}\{\|x\| \in C\}, \quad C \subset [0, \infty) \text{ Borel}.$$

Thus $\nu_K(C)$ is the expected number of Palm incident edges whose lengths belong to C .

Corollary 4.2 (Palm incident-edge radial intensity). *For a homogeneous Poisson input and a symmetric undirected unit-region rule, ν_K has density*

$$(4.3) \quad \nu_K(dr) = \rho d\kappa_d r^{d-1} \exp\{-\rho a_K r^d\} dr, \quad r \geq 0.$$

Proof. For every nonnegative measurable function $h : [0, \infty) \rightarrow [0, \infty)$, the Mecke–Slivnyak theorem identifies the reduced Palm process around the typical point with an ordinary homogeneous Poisson process. Therefore the Mecke equation gives

$$\mathbb{E}^0 \sum_{x \in \Phi \setminus \{0\}} \mathbf{1}\{0 \leftrightarrow x\} h(\|x\|) = \rho \int_{\mathbb{R}^d} h(\|x\|) \mathbb{P}\{\Phi(S(0, x)) = 0\} dx.$$

By the unit-region factorization,

$$\lambda_d(S(0, x)) = a_K \|x\|^d,$$

and hence the Poisson void probability gives

$$\mathbb{P}\{\Phi(S(0, x)) = 0\} = \exp\{-\rho a_K \|x\|^d\}.$$

Thus

$$\begin{aligned} \mathbb{E}^0 \sum_{x \in \Phi \setminus \{0\}} \mathbf{1}\{0 \leftrightarrow x\} h(\|x\|) &= \rho \int_{\mathbb{R}^d} h(\|x\|) \exp\{-\rho a_K \|x\|^d\} dx \\ &= \int_0^\infty h(r) \rho d\kappa_d r^{d-1} e^{-\rho a_K r^d} dr. \end{aligned}$$

This identifies the density in (4.3). □

Corollary 4.3 (Mean Palm degree and normalized radial law). *For a symmetric undirected unit-region rule, the expected Palm degree is*

$$(4.4) \quad \bar{d}_K := \mathbb{E}^0[\deg(0)] = \nu_K([0, \infty)) = \frac{\kappa_d}{a_K}.$$

After normalizing the first-moment measure ν_K by its total mass, the corresponding radial density is

$$(4.5) \quad g_K(r) = d\rho a_K r^{d-1} \exp\{-\rho a_K r^d\}, \quad r \geq 0.$$

Equivalently, if R_K has density g_K , then

$$(4.6) \quad \rho a_K R_K^d \sim \text{Exp}(1).$$

Consequently, for every $m > -d$,

$$(4.7) \quad \mathbb{E}[R_K^m] = (\rho a_K)^{-m/d} \Gamma\left(1 + \frac{m}{d}\right).$$

Proof. Integrating (4.3) and using the substitution $u = \rho a_K r^d$ gives

$$\nu_K([0, \infty)) = \rho d \kappa_d \int_0^\infty r^{d-1} e^{-\rho a_K r^d} dr = \frac{\kappa_d}{a_K}.$$

Dividing (4.3) by this total mass gives (4.5). The transformation $u = \rho a_K r^d$ also gives (4.6) and (4.7). $\square$

Corollary 4.4 (Order- k empty-region rule). *For a fixed deterministic candidate pair $p \neq q$, if the edge is retained when $S(p, q)$ contains at most $k \in \mathbb{N}_0$ Poisson points, then*

$$(4.8) \quad \mathbb{P}\{\Phi(S(p, q)) \leq k\} = e^{-\rho a_K \ell_{p,q}^d} \sum_{j=0}^k \frac{(\rho a_K \ell_{p,q}^d)^j}{j!}.$$

Under the two-point Palm interpretation, the same formula applies to the count of Poisson points other than the two endpoints.

Proof. By Theorem 3.3,

$$\rho \lambda_d(S(p, q)) = \rho a_K \ell_{p,q}^d.$$

The count $\Phi(S(p, q))$ is Poisson with this parameter, giving the Poisson distribution function in (4.8). $\square$

Quantity	Formula	Scope
Fixed-pair void probability	$e^{-\rho a_K r^d}$	fixed deterministic pair
Palm radial intensity	$\rho d \kappa_d r^{d-1} e^{-\rho a_K r^d} dr$	symmetric undirected rule
Mean Palm degree	$\mathbb{E}^0[\deg(0)] = \kappa_d / a_K$	symmetric undirected rule
Normalized Palm radial law	$\rho a_K R_K^d \sim \text{Exp}(1)$	normalized ν_K
Order- k void law	$e^{-\rho a_K r^d} \sum_{j=0}^k (\rho a_K r^d)^j / j!$	fixed pair; reduced Palm analogue

TABLE 1. Poisson quantities determined by the unit-volume constant a_K .

We next record the corresponding binomial local limit, followed by references where the same local scale appears and applications in which these templates are used.

5. BINOMIAL INPUTS AND EXPECTED-SIZE CONSTANTS

Let $X_1, \dots, X_n$ be independent $\mathbb{R}^d$ -valued random variables with common density f . Conditional on $X_i = p$ and $X_j = q$, the remaining $n - 2$ points are independent with the same density. Hence the exact probability that $S(p, q)$ contains no other sample point is

$$(5.1) \quad \begin{aligned} & \mathbb{P}\{S(p, q) \cap \{X_\ell : \ell \neq i, j\} = \emptyset \mid X_i = p, X_j = q\} \\ &= \left(1 - \int_{S(p,q)} f(u) du\right)^{n-2}. \end{aligned}$$

Assume now that S has a unit region K , so that

$$S(p, q) = p + \ell_{p,q} R_{p,q} K.$$

By the change of variables $u = p + \ell_{p,q} R_{p,q} z$,

$$(5.2) \quad \int_{S(p,q)} f(u) du = \ell_{p,q}^d \int_K f(p + \ell_{p,q} R_{p,q} z) dz.$$

Let x be an interior point at which f is continuous, and suppose that $p, q \rightarrow x$ and $\ell_{p,q} \rightarrow 0$. If K is bounded, then (5.2) gives

$$(5.3) \quad \int_{S(p,q)} f(u) \, du = f(x) a_K \ell_{p,q}^d + o(\ell_{p,q}^d).$$

The same expansion holds for finite-measure templates whenever

$$(5.4) \quad \int_K |f(p + \ell_{p,q} R_{p,q} z) - f(x)| \, dz \rightarrow 0.$$

This condition is automatic for bounded K under continuity of f at x .

In the local scaling

$$\ell_{p,q} = r n^{-1/d}, \quad r \geq 0,$$

equation (5.3) gives

$$(n-2) \int_{S(p,q)} f(u) \, du \rightarrow f(x) a_K r^d.$$

Therefore

$$(5.5) \quad \left(1 - \int_{S(p,q)} f(u) \, du \right)^{n-2} \rightarrow \exp\{-f(x) a_K r^d\}.$$

This is the binomial local analogue of the Poisson void factor, with local intensity $f(x)$.

The same calculation gives the local constant in expected-size asymptotics. Near an interior point x , candidate neighbors have local intensity $n f(x)$. A candidate at distance $r n^{-1/d}$ is accepted with approximate factor

$$\exp\{-f(x) a_K r^d\}.$$

Equivalently, for a typical point near x , the local expected number of accepted neighbors is approximated by

$$n \int_{\mathbb{R}^d} f(y) \exp\{-n f(x) a_K \|y - x\|^d\} \, dy.$$

With the change of variables $y = x + n^{-1/d} z$, this becomes

$$\int_{\mathbb{R}^d} f(x) \exp\{-f(x) a_K \|z\|^d\} \, dz = \frac{\kappa_d}{a_K}.$$

Thus the local mean degree constant is

$$(5.6) \quad \frac{\kappa_d}{a_K}.$$

For symmetric undirected rules, under the regularity and boundary hypotheses required for the corresponding expected-size theorem, the leading edge-count asymptotic has the form

$$(5.7) \quad \mathbb{E}|E_n| \sim \frac{n}{2} \frac{\kappa_d}{a_K}.$$

Devroye's results give density-independent lower bounds and exact constants for several sparse empty-region graphs, including Gabriel and relative-neighborhood graphs [5].

5.1. Applied uses of fixed-template proximity graphs. The same empty-region templates also appear outside the Poisson setting. In these applications the point set is usually observed, constructed, or simulated, and the graph is used as a geometric summary of adjacency, obstruction, or movement. The table records the role of the template in each setting; it is not an extension of the Poisson formulas above.

Setting	Role of the template	Representative source
Geographic variation and clustering	The Gabriel disk with diameter pq is used as a local adjacency criterion for geographic data.	Gabriel–Sokal; Matula–Sokal [7, 14]
Computational morphology and spatial analysis	Gabriel, relative-neighborhood, and β -skeleton regions encode neighborhood and shape information in finite planar point sets.	Toussaint; Jaromczyk–Toussaint [16, 9]
Transportation-network models	Proximity graphs are compared as idealized road-network skeletons and evaluated through travel-efficiency criteria.	Watanabe [18]
Street-network and tiling models	β -skeleton regions are used to recover cyclic tiling graphs motivated by street-network structure.	Maravillo Gómez–Calvillo Vives–Treviño Aguilar [22]
Public movement analysis	The stepping-stone region gives a parameterized diversion corridor between the Gabriel case and the relative-neighborhood case.	Kannangara–Tanin–Harwood–Karunasekera [10]
Network topology generation	Gabriel graph templates are used to generate reproducible reference physical network topologies.	Jurkiewicz [21]

TABLE 2. Applied settings where fixed-template proximity graphs are used as geometric adjacency rules.

6. EXAMPLES AND CONSTANTS

Graph or region	Unit region K for the normalized pair $(0, e_1)$	a_K
Source ball/contact region	$B(0, 1)$	κ_d
Gabriel region	$B(e_1/2, 1/2)$	$2^{-d}\kappa_d$
Relative-neighborhood lune	$B(0, 1) \cap B(e_1, 1)$	$\lambda_d(B(0, 1) \cap B(e_1, 1))$
Planar RNG	lens of two unit disks at distance 1	$2\pi/3 - \sqrt{3}/2$
Stepping-stone region	$\{z : \ z\ ^\alpha + \ z - e_1\ ^\alpha \leq 1\}$	$a_{d,SS}(\alpha)$, see (6.9)
Finite template ERG	any fixed finite-measure Borel template K	$\lambda_d(K)$

TABLE 3. Compact list of normalized templates and their volume constants.

6.1. Compact constants table.

6.2. **Balls, contact distance, and nearest-neighbor scale.** For the anchored source-ball rule,

$$S_{\text{ball}}(p, q) = B(p, \ell_{p,q}),$$

the unit region is $K = B(0, 1)$ and $a_K = \kappa_d$; see Figure 2.

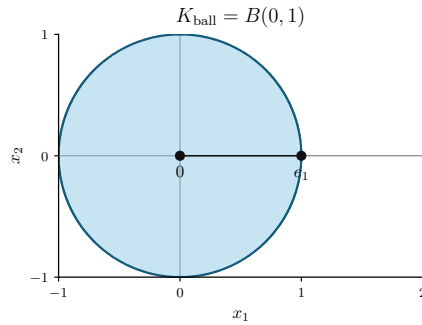


FIGURE 2. The normalized source-ball template $K_{\text{ball}} = B(0, 1)$. For the anchored rule $S_{\text{ball}}(p, q) = B(p, \ell_{p,q})$, the region is obtained from this unit ball by translation and scaling.

This yields the standard contact-distance scale

$$\mathbb{P}\{R > r\} = e^{-\rho\kappa_d r^d}, \quad \rho\kappa_d R^d \sim \text{Exp}(1).$$

The same empty-ball calculation is the local ingredient behind nearest-neighbor laws. For undirected nearest-neighbor graph conventions, the degree formula depends on whether the graph is directed, mutual, or symmetrized.

6.3. Gabriel and relative-neighborhood regions. Figure 3 shows the two normalized templates used in this subsection.

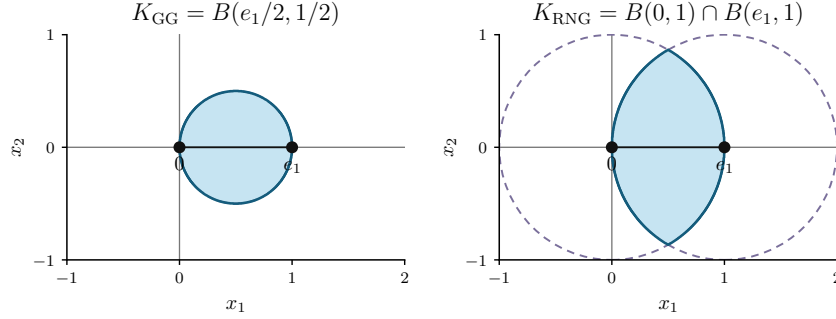


FIGURE 3. Normalized Gabriel and relative-neighborhood templates. In the normalized pair $(0, e_1)$, the Gabriel template is the ball with diameter $[0, e_1]$, whereas the relative-neighborhood template is $B(0, 1) \cap B(e_1, 1)$.

For the Gabriel graph,

$$S_{\text{GG}}(p, q) = B\left(\frac{p+q}{2}, \frac{\ell_{p,q}}{2}\right), \quad K_{\text{GG}} = B\left(\frac{e_1}{2}, \frac{1}{2}\right),$$

so

$$(6.1) \quad a_{d,\text{GG}} = 2^{-d}\kappa_d.$$

In the plane, $a_{2,\text{GG}} = \pi/4$, hence the Poisson Palm mean degree is $\kappa_2/a_{2,\text{GG}} = 4$ and the expected edge count has leading term $2n$.

For the relative-neighborhood graph,

$$S_{\text{RNG}}(p, q) = B(p, \ell_{p,q}) \cap B(q, \ell_{p,q}),$$

so

$$(6.2) \quad a_{d,\text{RNG}} = \lambda_d(B(0, 1) \cap B(e_1, 1)).$$

Equivalently,

$$(6.3) \quad a_{d,\text{RNG}} = \kappa_d I_{3/4}\left(\frac{d+1}{2}, \frac{1}{2}\right),$$

where $I_x(a, b)$ is the regularized incomplete beta function. For $d = 2$,

$$a_{2,\text{RNG}} = \frac{2\pi}{3} - \frac{\sqrt{3}}{2}.$$

This gives the planar Palm mean degree $\pi/(2\pi/3 - \sqrt{3}/2)$ and the leading expected edge count $(n/2)\pi/(2\pi/3 - \sqrt{3}/2)$.

6.4. β -skeletons, finite Cardinal–Collette–Langerman templates, and finite Veltkamp regions. For each fixed parameter in the usual lune-based β -skeleton family, the region is obtained by scaling and rotating a normalized lune; Gabriel and RNG occur as special cases in standard conventions [11, 4]. Thus the unit-region theorem applies with $K = K_\beta$ and $a_K = \lambda_d(K_\beta)$. Figure 4 displays a representative β -skeleton template and two finite Veltkamp-type ball templates.

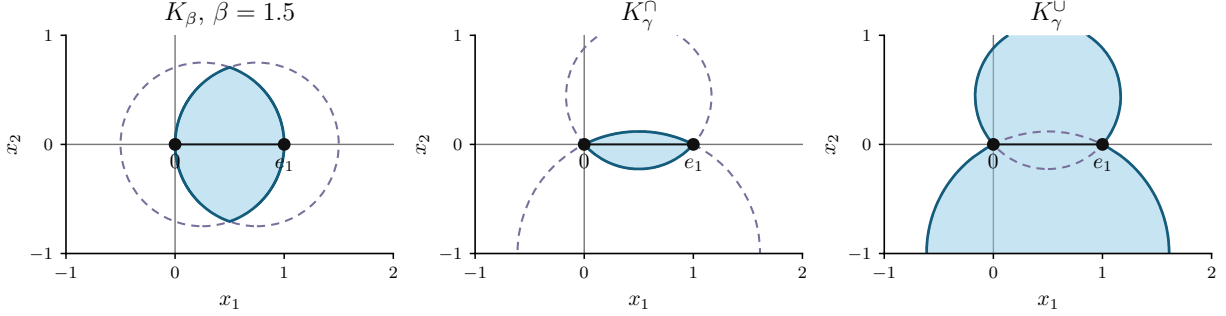


FIGURE 4. Finite ball-generated templates. The left panel shows a representative β -skeleton lune. The middle and right panels show two-ball intersection and union templates in normalized coordinates. The cap-volume formula below is used for the two-ball intersection case.

Cardinal–Collette–Langerman’s finite empty-region templates satisfy the same criterion by construction: the region associated with a pair is obtained from a fixed template by translation, rotation, and uniform scaling [4]. Infinite regions, such as strips or halfspaces, may still be similarity templates with $a_K = \infty$, placing them outside the finite-volume Poisson formulas above.

Veltkamp’s γ -neighborhood graph unifies several geometric graphs by using regions built from balls, spheres, and limiting halfspaces [17]. Whenever the chosen γ -region is finite and built from finitely many balls whose centers are affine functions of (p, q) and whose radii are proportional to $\ell_{p,q}$, it has a finite unit region. Its constant is a finite union/intersection volume of unit balls. In the two-ball case this constant has an explicit special-function form. Let

$$K = B(c_0, R_0) \cap B(c_1, R_1), \quad \delta = \|c_1 - c_0\|,$$

with $R_0, R_1 > 0$. If $\delta \geq R_0 + R_1$, then $a_K = 0$. If $\delta \leq |R_0 - R_1|$, then the smaller ball is contained in the larger one and

$$a_K = \kappa_d \min\{R_0, R_1\}^d.$$

For the transverse overlap case

$$|R_0 - R_1| < \delta < R_0 + R_1,$$

define

$$x_0 = \frac{\delta^2 + R_0^2 - R_1^2}{2\delta}, \quad x_1 = \delta - x_0,$$

and the corresponding cap heights

$$h_0 = R_0 - x_0, \quad h_1 = R_1 - x_1.$$

Then the unit volume is the sum of two hyperspherical caps,

$$(6.4) \quad a_K = V_{\text{cap}}^{(d)}(R_0, h_0) + V_{\text{cap}}^{(d)}(R_1, h_1).$$

Following Li’s hyperspherical-cap formula [13], for $0 \leq h \leq 2R$,

$$(6.5) \quad V_{\text{cap}}^{(d)}(R, h) = \begin{cases} \frac{1}{2} \kappa_d R^d I_{1-(1-h/R)^2} \left(\frac{d+1}{2}, \frac{1}{2} \right), & 0 \leq h \leq R, \\ \kappa_d R^d - \frac{1}{2} \kappa_d R^d I_{1-(1-h/R)^2} \left(\frac{d+1}{2}, \frac{1}{2} \right), & R \leq h \leq 2R, \end{cases}$$

where $I_x(a, b)$ is the regularized incomplete beta function. Combining (6.4) and (6.5) gives a closed expression for every finite two-ball intersection template; finite unions are obtained by inclusion–exclusion. For the relative-neighborhood lune, $c_0 = 0$, $c_1 = e_1$, and $R_0 = R_1 = 1$, so $x_0 = x_1 = 1/2$ and $h_0 = h_1 = 1/2$. Thus

$$a_{d,\text{RNG}} = \kappa_d I_{3/4} \left(\frac{d+1}{2}, \frac{1}{2} \right),$$

which is the constant in (6.3).

6.5. Stepping-stone diversion regions. For $\alpha \geq 1$, define the stepping-stone region

$$(6.6) \quad S_{\text{SS},\alpha}(p, q) = \{z \in \mathbb{R}^d : \|z - p\|^\alpha + \|z - q\|^\alpha \leq \ell_{p,q}^\alpha\}.$$

It has unit representative

$$K_{\text{SS},\alpha} = \{z \in \mathbb{R}^d : \|z\|^\alpha + \|z - e_1\|^\alpha \leq 1\}, \quad a_{d,\text{SS}}(\alpha) = \lambda_d(K_{\text{SS},\alpha}).$$

For $d = 1$, the unit region is the interval $K_{\text{SS},\alpha} = [0, 1]$, so

$$a_{1,\text{SS}}(\alpha) = 1, \quad \alpha \geq 1.$$

For $\alpha = 1$ and $d \geq 2$, the triangle inequality forces equality on the segment $[p, q]$, whose d -dimensional volume is zero. For $d \geq 2$ and $\alpha > 1$, the constant has a one-dimensional integral representation. Set

$$(6.7) \quad y_\alpha(u) = \frac{1}{2} \left[4u^2 - (1 + u^2 - (1 - u^\alpha)^{2/\alpha})^2 \right]^{1/2}$$

and

$$(6.8) \quad x'_\alpha(u) = u + u^{\alpha-1}(1 - u^\alpha)^{(2-\alpha)/\alpha}.$$

Then

$$(6.9) \quad a_{d,\text{SS}}(\alpha) = 2\kappa_{d-1} \int_{2^{-1/\alpha}}^1 y_\alpha(u)^{d-1} x'_\alpha(u) du.$$

Consequently,

$$(6.10) \quad \lambda_d(S_{\text{SS},\alpha}(p, q)) = \ell_{p,q}^d a_{d,\text{SS}}(\alpha).$$

A detailed derivation of the integral formula (6.9) is given in the companion manuscript [6]; here it is used only as the unit-volume constant for the stepping-stone template.

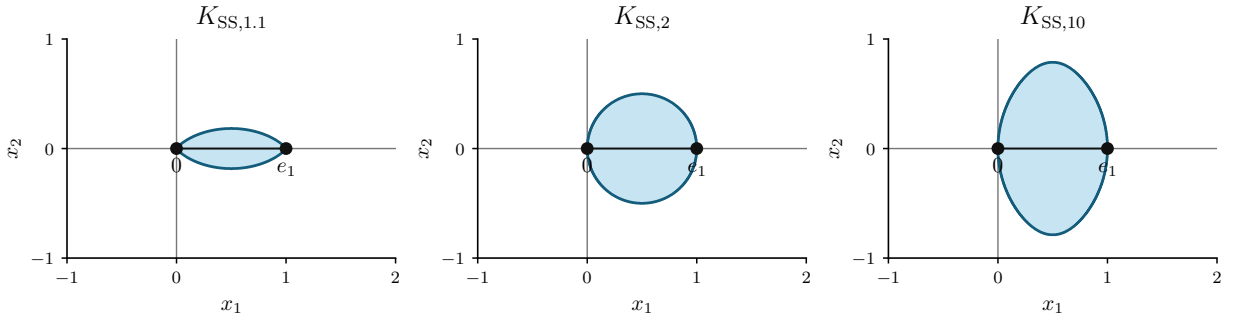


FIGURE 5. Normalized stepping-stone templates for the pair $(0, e_1)$. The case $\alpha = 2$ is the Gabriel template. As α increases, the templates approach the relative-neighborhood lune.

The main anchor cases agree with the usual proximity regions. When $\alpha = 2$, the parallelogram identity gives

$$\|z - p\|^2 + \|z - q\|^2 = 2 \|z - (p + q)/2\|^2 + \frac{\ell_{p,q}^2}{2},$$

so $S_{\text{SS},2}(p, q)$ is the Gabriel ball of radius $\ell_{p,q}/2$ and

$$a_{d,\text{SS}}(2) = 2^{-d} \kappa_d.$$

As $\alpha \rightarrow \infty$,

$$(a^\alpha + b^\alpha)^{1/\alpha} \rightarrow \max\{a, b\}, \quad a, b \geq 0,$$

so the limiting region is

$$\{z : \max\{\|z - p\|, \|z - q\|\} \leq \ell_{p,q}\} = B(p, \ell_{p,q}) \cap B(q, \ell_{p,q}).$$

The limiting constant is therefore the relative-neighborhood constant

$$\lim_{\alpha \rightarrow \infty} a_{d,SS}(\alpha) = \kappa_d I_{3/4} \left(\frac{d+1}{2}, \frac{1}{2} \right).$$

The stepping-stone graph was introduced for movement analysis [10]; the companion manuscript [6] gives the detailed slicing proof of (6.9).

7. SCOPE OF THE RESULT

The unit-region theorem applies to rules in which each candidate pair $p \neq q$ is assigned one prescribed finite-measure Borel region $S(p, q)$, and this region is a similarity copy of a fixed template K . Thus the result is a local pairwise statement. This is the setting of fixed empty-region graphs in the sense of Devroye and of finite template regions in the sense of Cardinal–Collette–Langerman [5, 4]. Constructions whose edge rules involve an existential choice, all sites at once, a global optimization problem, an infinite region, or an external orientation convention require separate treatment.

- **Delaunay graph.** Delaunay adjacency is an empty-ball condition, but it is existential: an edge exists when there is some empty ball whose boundary contains p and q . Thus the test is over a family of balls rather than over one prescribed region $S(p, q)$. Devroye makes the same distinction when noting that the Delaunay triangulation is not a special case of his fixed- $S(X_i, X_j)$ framework [5]. The Delaunay graph remains closely related to the present examples through the classical inclusion

$$\text{MST} \subseteq \text{RNG} \subseteq \text{GG} \subseteq \text{DT},$$

recorded in the proximity-graph literature [5, 16, 9].

- **Voronoi cells.** A Voronoi cell is defined by comparing one site with every other site. Its geometry is an all-site construction, not a pairwise empty-region rule. Voronoi and Delaunay structures are central objects in computational geometry and are also part of the broader neighborhood-graph setting discussed by Veltkamp [17]. Volume, face, and stabilization questions for Voronoi tessellations therefore belong to the theory of geometric functionals rather than to the fixed-template calculation used here [15, 12].
- **Minimum spanning tree.** The minimum spanning tree is selected by a global optimization criterion. In the plane it is related to the relative-neighborhood and Gabriel graphs through the standard inclusions above, but its edge set is not obtained by testing one empty region for each candidate pair independently [16, 9, 5].
- **Infinite strips and halfspaces.** Some similarity-covariant regions have infinite Lebesgue measure. For such templates $a_K = \lambda_d(K) = \infty$, so the finite-volume Poisson formulas in this paper do not apply. Devroye discusses the infinite-strip graph separately and notes that it requires different tools, while Veltkamp treats halfspaces as limiting neighborhoods in the γ -neighborhood setting [5, 17]. Such regions may still define deterministic proximity rules, but their stochastic analysis needs different normalization or localization.
- **Nearest-neighbor variants.** The source-ball rule gives the usual nearest-neighbor distance scale, but graph conventions matter. Directed, mutual, and symmetrized nearest-neighbor graphs have different degree formulas. Devroye treats nearest-neighbor graphs as a separate example, and Veltkamp includes nearest-neighbor graphs among the basic geometric neighborhood graphs [5, 17]. Thus the empty-ball volume calculation supplies the local scale, while the graph-level interpretation depends on the chosen convention.
- **Standard Θ -graphs and Yao-type cone graphs.** Standard cone graphs use cones attached to an external coordinate frame, usually at each source point. This convention breaks the Euclidean same-template covariance used in Theorem 3.3: after rotating the whole configuration, the prescribed cones do not rotate with each candidate pair unless that convention is built into the rule. These graphs are better treated as coordinate-anchored cone constructions, as in the Yao and Θ -graph literature [1, 2, 3].

A rotating-cone variant can be made pairwise-covariant, but that is a different rule from the standard coordinate-anchored construction.

8. CONCLUSION

For an empty-region rule with a unit region,

$$S(p, q) = p + \ell_{p,q} R_{p,q} K, \quad a_K = \lambda_d(K),$$

the volume identity

$$\lambda_d(S(p, q)) = a_K \ell_{p,q}^d$$

follows directly from translation invariance, orthogonal invariance, and scaling of Lebesgue measure. This elementary identity is the common geometric input behind the fixed-pair Poisson void law, the order- k extension, and the local binomial scaling. For symmetric undirected rules it also gives the Palm incident-edge radial intensity, the normalized Palm edge-length distribution, and the mean Palm degree.

The examples above show that several standard proximity regions are governed by the same mechanism: once the pair (p, q) is normalized to $(0, e_1)$, the stochastic formulas depend on the graph only through the unit-template volume a_K . Gabriel regions, relative-neighborhood lunes, fixed-parameter β -skeleton regions, finite Veltkamp-type ball regions, and stepping-stone regions differ geometrically, but their homogeneous Poisson void scale has the same form,

$$\rho \lambda_d(S(p, q)) = \rho a_K \ell_{p,q}^d.$$

Further asymptotic results, such as expected-size limits, stabilization theorems, normal approximations, and longest-edge laws, use this same local scale together with their own assumptions.

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